

LECTURE-49 PROPERTIES OF LAPLACE TRANSFORM PART-02



Question-13

Determine Laplace transform of following signals

$$(a) \quad x(t) = \frac{\sin t}{t} u(t)$$

$$(b) \quad x(t) = \left\{ \frac{e^{-at} - e^{-bt}}{t} \right\} u(t)$$

$$X(s) = \ln \left(\frac{s+b}{s+a} \right)$$

$$(b) \quad e^{-at} u(t) \rightarrow \frac{1}{s+a}$$

$$e^{-bt} u(t) \rightarrow \frac{1}{s+b}$$

$$x(t) \rightarrow \int_s^\infty \left(\frac{1}{\lambda+a} - \frac{1}{\lambda+b} \right) d\lambda$$

$$= \left[\ln(\lambda+a) - \ln(\lambda+b) \right]_s^\infty$$

$$= \left[\ln \left(\frac{\lambda+a}{\lambda+b} \right) \right]_s^\infty$$

$$= \left[\lim_{\lambda \rightarrow \infty} \ln \left(\frac{\lambda+a}{\lambda+b} \right) - \ln \left(\frac{s+a}{s+b} \right) \right]$$

Question-14

If $x(t) = 2\sqrt{\frac{t}{\pi}} \rightarrow s^{-3/2}$. Then find Laplace transform of $\sqrt{\frac{1}{\pi t}}$?

$$2\sqrt{\frac{t}{\pi}} \rightarrow s^{-3/2}$$

$$\frac{2}{t} \sqrt{\frac{t}{\pi}} = \frac{2}{\sqrt{\pi t}} \rightarrow \int_s^\infty \lambda^{-3/2} d\lambda = \left[-\frac{\lambda^{-1/2}}{1/2} \right]_s^\infty$$

$$\frac{2}{\sqrt{\pi t}} \rightarrow \frac{2}{\sqrt{s}}$$

$$\frac{1}{\sqrt{\pi t}} \rightarrow \frac{1}{\sqrt{s}}$$

Conjugation in time domain

• if $x(t) \xrightarrow{LT} X(s)$ ROC = R

then $x^*(t) \xrightarrow{LT} X^*(s^*)$ ROC : $\text{Re}\{s^*\}$: same

proof

$$y(t) = x^*(t)$$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

$$Y(s) = \int_{-\infty}^{\infty} x^*(t) e^{-st} dt$$

$$Y^*(s) = \int_{-\infty}^{\infty} x(t) e^{-s^*t} dt$$

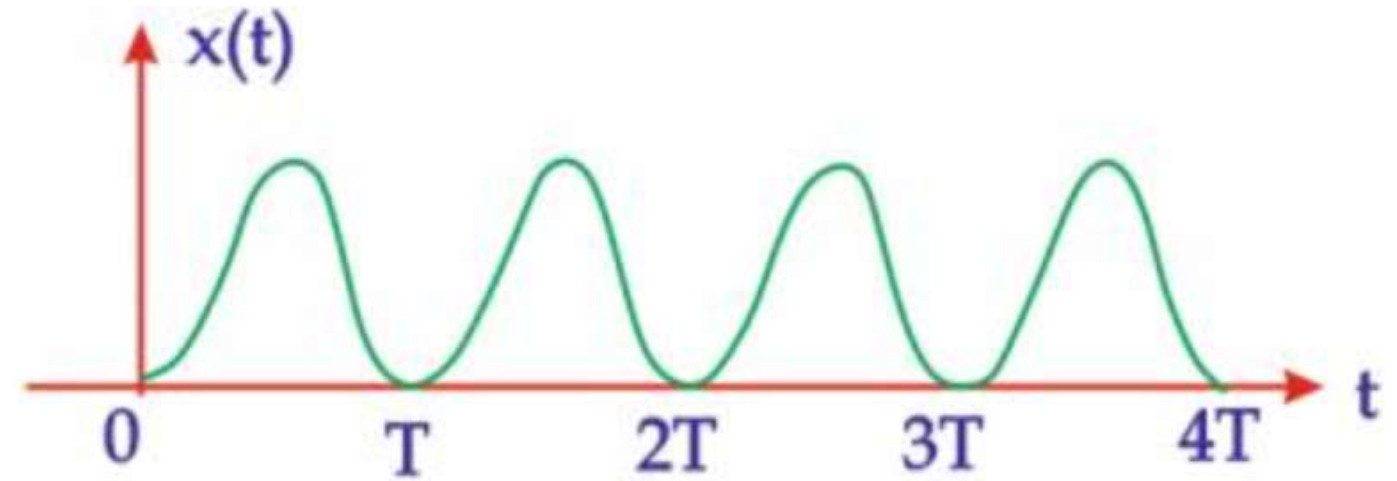
$$Y^*(s) = X(s^*)$$

$$Y(s) = X^*(s^*)$$

Bilateral Laplace Transform of Causal Periodic Signals

**
SN

- BLT cannot be determined for periodic signals extending from $-\infty$ to ∞ but we can determine for causal periodic signal or semi-periodic signal $t \in (0, \infty)$



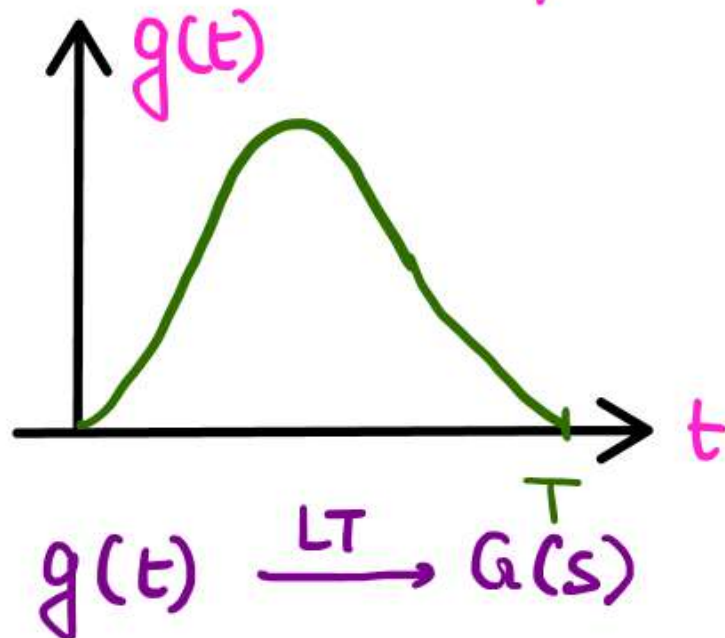
$$x(t) = g(t) + g(t-T) + g(t-2T) + \dots$$

$$X(s) = G(s) + e^{-sT} G(s) + e^{-2sT} G(s) + \dots$$

$$X(s) = G(s) [1 + e^{-sT} + e^{-2sT} + \dots]$$

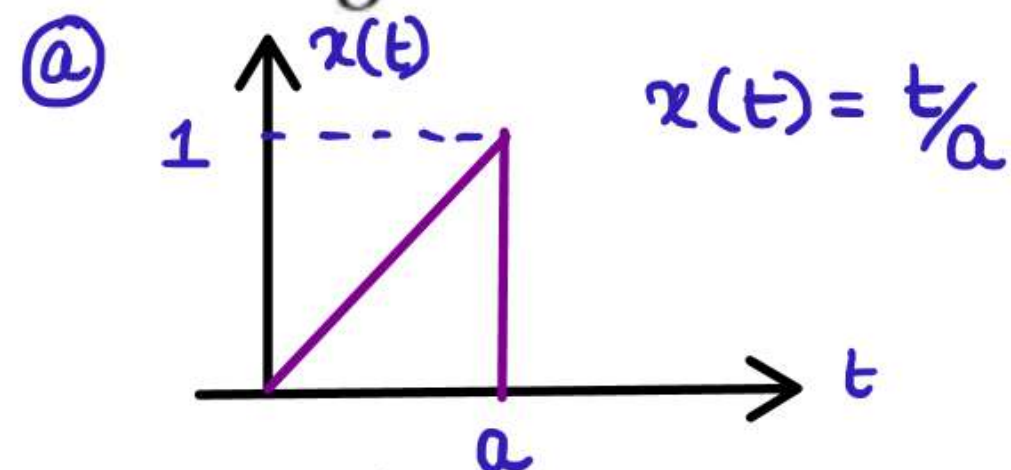
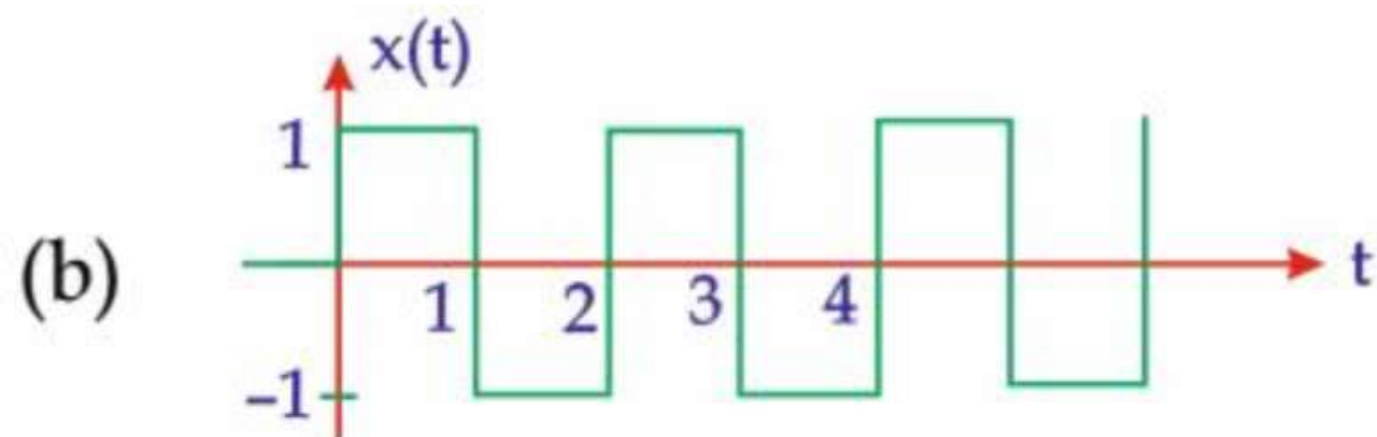
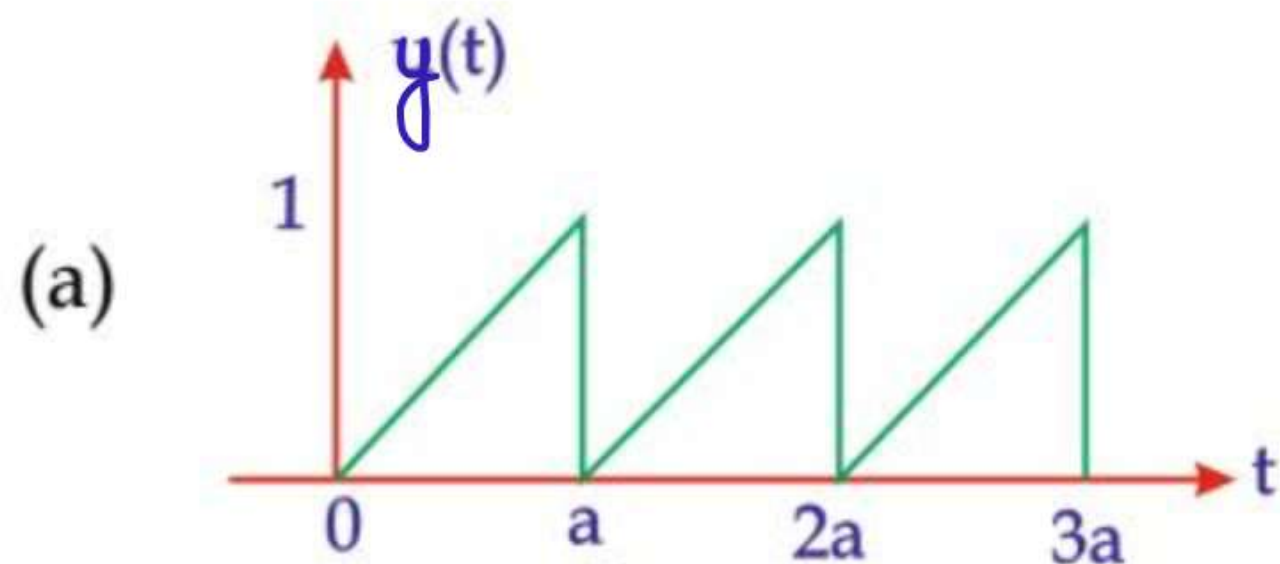
$$X(s) = \frac{G(s)}{1 - e^{-sT}}$$

Consider one period of $x(t)$



Question-01

Determine Laplace Transform of following periodic signals



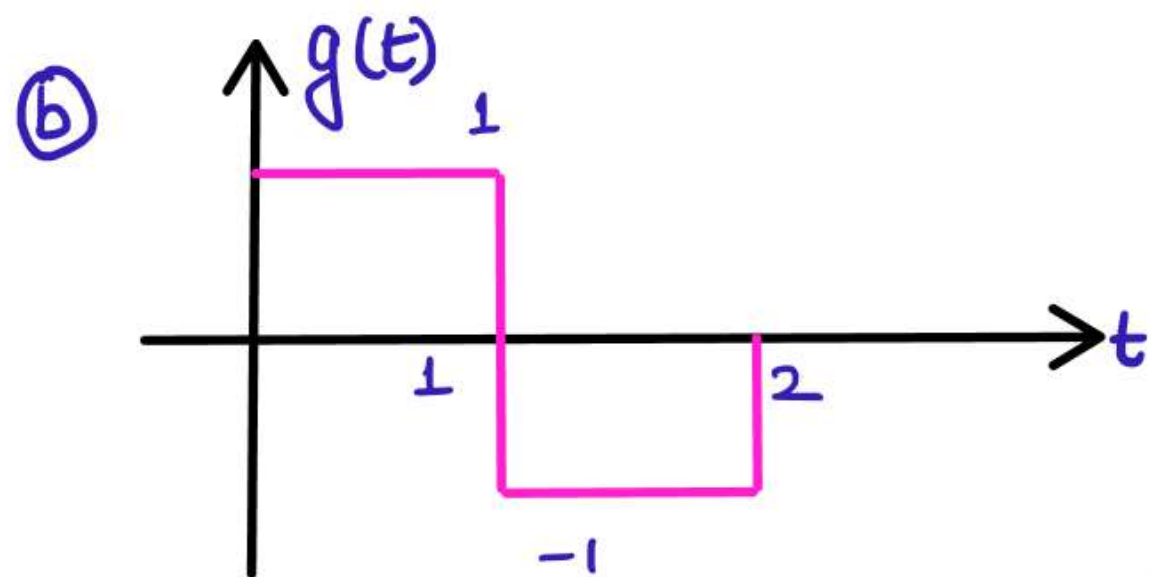
$$X(s) = \int_0^a \frac{t}{a} e^{-st} dt$$

$$= \frac{1}{a} \left[t \frac{e^{-st}}{-s} - \frac{e^{-st}}{s^2} \right]_0^a$$

$$= \frac{1}{a} \left[\frac{ae^{-as}}{-s} - \frac{e^{-as}}{s^2} + \frac{1}{s^2} \right]$$

$$X(s) = \frac{1}{as^2} [1 - e^{-as} - as e^{-as}]$$

$$Y(s) = \frac{X(s)}{1 - e^{-s(a)}} = \frac{X(s)}{1 - e^{-as}} = \frac{1}{as^2} - \frac{e^{-as}}{s(1 - e^{-as})}$$



$$\begin{aligned} G(s) &= \int_0^1 1 e^{-st} dt + \int_1^2 -1 e^{-st} dt \\ &= \left[\frac{e^{-st}}{-s} \right]_0^1 - \left[\frac{e^{-st}}{-s} \right]_1^2 \end{aligned}$$

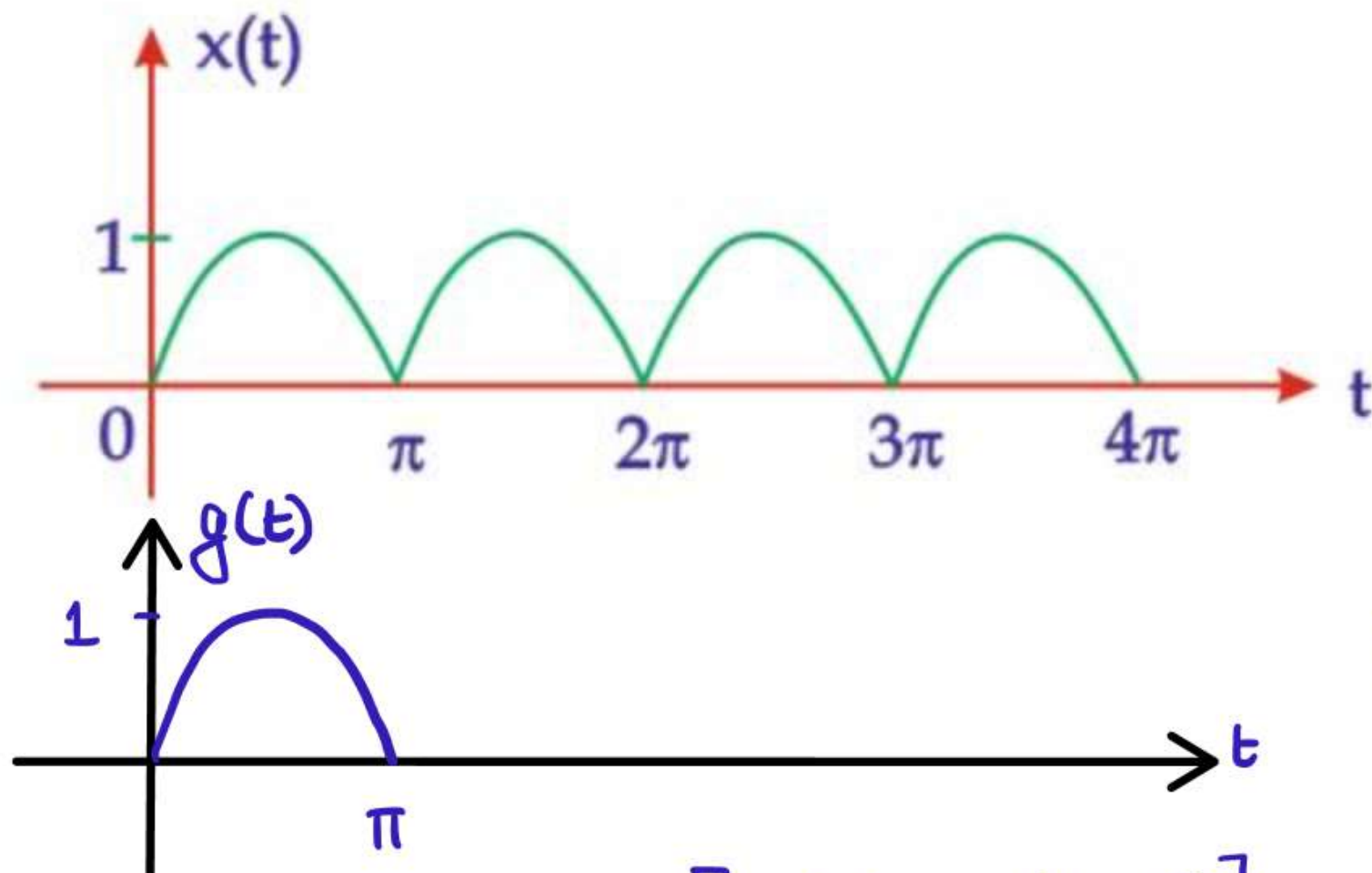
$$\begin{aligned} G(s) &= \frac{1 - e^{-s}}{s} + \frac{e^{-2s} - e^{-s}}{s} \\ &= \left(\frac{1 - 2e^{-s} + e^{-2s}}{s} \right) = \left(\frac{1 - e^{-s}}{s} \right)^2 \end{aligned}$$

$$\begin{aligned} X(s) &= \frac{G(s)}{1 - e^{-2s}} = \frac{G(s)}{(1 + e^{-s})(1 - e^{-s})} \\ &= \frac{(1 - e^{-s})}{s(1 + e^{-s})} \end{aligned}$$

Question-02

Determine Laplace Transform of following periodic signals

(a)



$$g(t) = \sin \pi t [u(t) - u(t - \pi)]$$

$$g(t) = \sin \pi t u(t) + \sin(t - \pi) u(t - \pi)$$

$$G(s) = \frac{(1 + e^{-s\pi})}{s^2 + 1}$$

$$X(s) = \frac{G(s)}{1 - e^{-s\pi}} = \frac{(1 + e^{-s\pi})}{(1 - e^{-s\pi})(s^2 + 1)}$$

$$X(s) = \frac{e^{-s\pi/2} (e^{s\pi/2} + e^{-s\pi/2})}{e^{-s\pi/2} (e^{s\pi/2} - e^{-s\pi/2}) (s^2 + 1)}$$

$$= \frac{2 \cosh(s\pi/2)}{2 \sinh(s\pi/2)} \times \frac{1}{s^2 + 1}$$

$$= \coth(s\pi/2) \cdot \frac{1}{s^2 + 1}$$

Inverse Laplace Transform

Synthesis equation

$$x(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} X(s) e^{st} ds$$

$\sigma \in \text{ROC}$

For rational function, $X(s) = \frac{N(s)}{D(s)}$

- ① compute poles of $x(s)$ $\{ D(s)=0 \}$
- ② consider ROC
- ③ find coefficient of each term corresponding to each pole by partial fraction.

④ Based on ROC, compute inverse LT

Partial fraction Expansion

- required when we wish to express a complicated fraction as sum of simpler fractions.

Conventional Method

$$\begin{aligned} F(s) &= \frac{s+3}{s^3 + 7s^2 + 10s} \\ &= \frac{(s+3)}{s(s+2)(s+5)} \\ &= \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+5} \end{aligned}$$

$$\begin{aligned} A &= \lim_{s \rightarrow 0} s F(s) = \lim_{s \rightarrow 0} \frac{(s+3)}{(s+2)(s+5)} \\ &= 3/10 \end{aligned}$$

$$\begin{aligned} B &= \lim_{s \rightarrow -2} (s+2) F(s) = \lim_{s \rightarrow -2} \frac{(s+3)}{s(s+5)} \\ &= \frac{1}{(-2)(-2+5)} = -1/6 \end{aligned}$$

$$\begin{aligned} C &= \lim_{s \rightarrow -5} (s+5) F(s) = \lim_{s \rightarrow -5} \frac{(s+3)}{s(s+2)} \\ &= \frac{-5+3}{-5(-5+2)} = -2/15 \end{aligned}$$

$$F(s) = \frac{3}{10s} - \frac{1}{6(s+2)} - \frac{2}{15(s+5)}$$

Conventional method

Partial fraction expansion can only be performed if order of num^r < order of deno^r i.e proper function $X(s)$.

if order of num^r $\geq$ order of deno^r then we first perform division of 2 polynomial & then use PF.

eg: $F(s) = \frac{6s^2 + 3s + 2}{2s^2 + 14s + 20}$

$$\begin{array}{r} 2s^2 + 14s + 20 \overline{) 6s^2 + 3s + 2} \quad 3 \\ \underline{6s^2 + 42s + 60} \\ -39s - 58 \end{array}$$

$$3 - \frac{(39s + 58)}{2s^2 + 14s + 20}$$

PF can be applied

Case-01: Distinct real roots

$$F(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{\prod_{i=1}^n (s+a_i)}$$

$$= \sum_{i=1}^n \frac{A_i}{s+a_i}$$

$$A_i = \lim_{s \rightarrow -a_i} (s+a_i) F(s)$$

Cover-up method

$$F(s) = \frac{s+3}{s(s+2)(s+5)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+5}$$

$$A = \lim_{s \rightarrow 0} \frac{s+3}{s(s+2)(s+5)} = 3/10$$

$$B = \lim_{s \rightarrow -2} \frac{s+3}{s(s+2)(s+5)} = -1/6$$

$$C = \lim_{s \rightarrow -5} \frac{s+3}{s(s+2)(s+5)} = -2/15$$

Case-02: repeated real roots

$$F(s) = \frac{s+3}{s(s+2)^2(s+5)}$$

$$= \frac{A}{s} + \frac{B}{s+2} + \frac{C}{(s+2)^2} + \frac{D}{s+5}$$

$$A = \lim_{s \rightarrow 0} \frac{s+3}{s(s+2)^2(s+5)} = 3/20$$

$$C = \lim_{s \rightarrow -2} \frac{s+3}{s(s+2)^2(s+5)} = -1/6$$

$$D = \lim_{s \rightarrow -5} \frac{s+3}{s(s+2)^2(s+5)} = 2/45$$

$$B = \lim_{s \rightarrow -2} \frac{d}{ds} (s+2)^2 F(s)$$

$$= \lim_{s \rightarrow -2} \frac{d}{ds} \left[\frac{s+3}{s(s+5)} \right]$$

$$= \lim_{s \rightarrow -2} \left[\frac{(s^2+5s) - (s+3)(2s+5)}{(s^2+5s)^2} \right]$$

$$= \frac{4-10 - (1)(1)}{(4-10)^2}$$

$$= -7/36$$

$$\text{eg: } \frac{1}{s(s+1)^3}$$

$$A/s + B/s+1 + C/(s+1)^2 + D/(s+1)^3$$

$$D = \lim_{s \rightarrow -1} (s+1)^3 F(s)$$

$$C = \lim_{s \rightarrow -1} \frac{d}{ds} ((s+1)^3 F(s))$$

$$B = \frac{1}{2!} \lim_{s \rightarrow -1} \frac{d^2}{ds^2} ((s+1)^3 F(s))$$

Case-03: complex roots

$$F(s) = \frac{(s+3)}{(s+5)(s^2+4s+5)}$$
$$= \frac{A}{s+5} + \frac{Bs+C}{s^2+4s+5}$$

$$A = \lim_{s \rightarrow -5} (s+5) F(s)$$
$$= \lim_{s \rightarrow -5} \frac{(s+3)}{(s^2+4s+5)}$$
$$= \frac{-2}{25-20+5} = -\frac{2}{10}$$

$$F(s) = \frac{-0.2}{s+5} + \frac{Bs+C}{s^2+4s+5}$$
$$= \frac{s^2(B-0.2) + s(5B+C-0.8) + (5C-1)}{(s+5)(s^2+4s+5)}$$

$$B-0.2=0$$

$$B=0.2$$

$$5C-1=3$$

$$C=0.8$$

Question-03

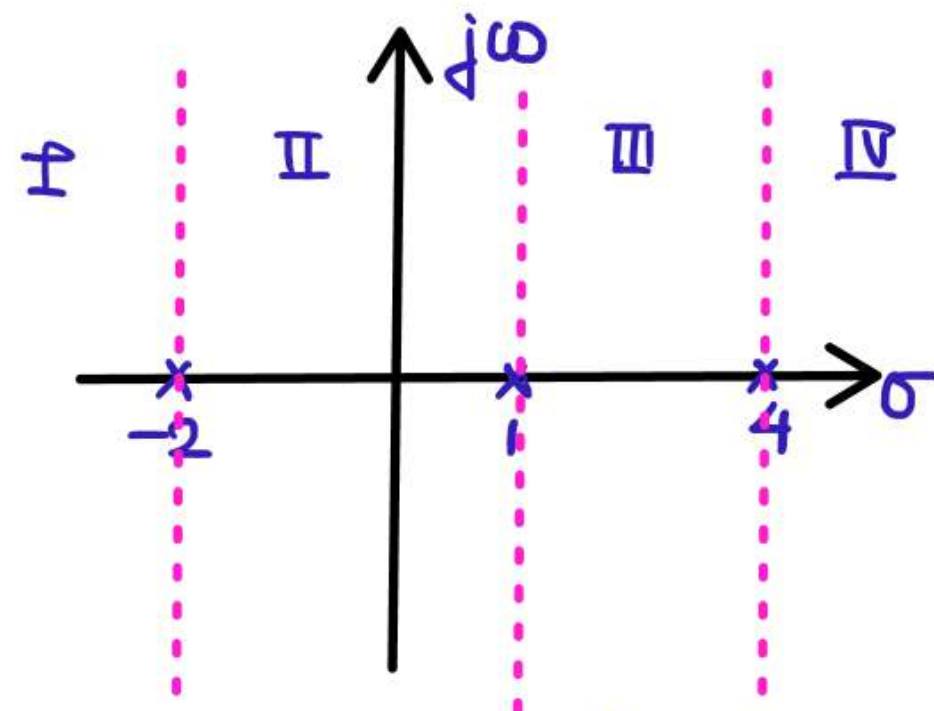
$$X(s) = \frac{1}{(s-1)(s+2)(s-4)^2} \text{ find } x(t)?$$

$$= \frac{A}{s-1} + \frac{B}{s+2} + \frac{C}{s-4} + \frac{D}{(s-4)^2}$$

$$A = \lim_{s \rightarrow 1} \frac{1}{(s-1)(s+2)(s-4)^2} = \frac{1}{27}$$

$$B = \lim_{s \rightarrow -2} \frac{1}{(s-1)(s+2)(s-4)^2} = -\frac{1}{108}$$

$$D = \lim_{s \rightarrow 4} \frac{1}{(s-1)(s+2)(s-4)^2} = \frac{1}{18}$$



$$C = \lim_{s \rightarrow 4} \frac{d}{ds} (s-4)^2 X(s)$$

$$= \lim_{s \rightarrow 4} \frac{d}{ds} \left[\frac{1}{(s-1)(s+2)} \right]$$

$$= \lim_{s \rightarrow 4} \frac{-(2s+1)}{(s^2+s-2)^2} = -\frac{1}{36}$$

$$e^{-at} u(t) \rightarrow \frac{1}{s+a} \quad \sigma > -a$$

$$-e^{-at} u(t) \rightarrow \frac{1}{s+a} \quad \sigma < -a$$

Case-01: $\sigma < -2$ (left of all poles)

$$X(s) = \frac{1}{27(s-1)} - \frac{1}{108(s+2)} - \frac{1}{36(s-4)} + \frac{1}{18(s-4)^2}$$

$$x(t) = -\frac{1}{27} e^t u(-t) + \frac{1}{108} e^{-2t} u(-t) + \frac{1}{36} e^{4t} u(-t) - \frac{t}{18} e^{4t} u(-t)$$

Case-02: $-2 < \sigma < 1$

$$x(t) = -\frac{1}{27} e^t u(-t) - \frac{1}{108} e^{-2t} u(t) + \frac{1}{36} e^{4t} u(-t) - \frac{t}{18} e^{4t} u(-t)$$

Case-03: $1 < \sigma < 4$

$$x(t) = \frac{1}{27} e^t u(t) - \frac{1}{108} e^{-2t} u(t) + \frac{1}{36} e^{4t} u(-t) - \frac{t}{18} e^{4t} u(-t)$$

Case-04: $\sigma > 4$

$$x(t) = \frac{1}{27} e^t u(t) - \frac{1}{108} e^{-2t} u(t) - \frac{1}{36} e^{4t} u(t) + \frac{t}{18} e^{4t} u(t)$$

ROC for absolutely integrable signal

ROC contains all such values of σ
s.t. $x(t)e^{-\sigma t}$ is absolutely integrable

$$\int_{-\infty}^{\infty} |x(t)e^{-\sigma t}| dt = \text{finite}$$

If signal is absolutely integrable

$$\int_{-\infty}^{\infty} |x(t)| dt = \text{finite}$$

Comparing 2 equⁿ we can say
 $\sigma=0$ is a part of ROC

****SN**

- A signal is absolutely integrable if ROC contains $\sigma=0$ or imaginary axis.

Question-04

Find the Inverse Laplace transform of

$$F(s) = \frac{s^2 - 4s + 7}{(s+2)^3}$$

$$= \frac{A}{(s+2)} + \frac{B}{(s+2)^2} + \frac{C}{(s+2)^3}$$

$$C = \lim_{s \rightarrow -2} (s+2)^3 F(s)$$

$$= \lim_{s \rightarrow -2} (s^2 - 4s + 7) = 19$$

$$B = \lim_{s \rightarrow -2} \frac{d}{ds} (s+2)^2 F(s)$$

$$B = \lim_{s \rightarrow -2} \frac{d}{ds} (s^2 - 4s + 7) = -8$$

$$A = \frac{1}{2!} \lim_{s \rightarrow -2} \frac{d^2}{ds^2} (s^2 - 4s + 7) = 1$$

$$F(s) = \frac{1}{s+2} - \frac{8}{(s+2)^2} + \frac{19}{(s+2)^3}$$

$$f(t) = \left[1 - 8t + \frac{19}{2} t^2 \right] e^{-2t} u(t)$$

$$t^2 e^{-2t} u(t) \rightarrow \frac{2}{(s+2)^3}$$

Question-05

Find the inverse Laplace transform of

$$F(s) = \frac{6 + (s+1)(s^2 - 5s + 11)}{s(s-1)(s-2)(s+1)} = \frac{A}{s} + \frac{B}{(s-1)} + \frac{C}{(s-2)} + \frac{D}{(s+1)}$$

$$A = \lim_{s \rightarrow 0} \frac{6 + (s+1)(s^2 - 5s + 11)}{s(s-1)(s-2)(s+1)} = \frac{17}{2}$$

$$D = \lim_{s \rightarrow -1} \frac{6 + (s+1)(s^2 - 5s + 11)}{s(s-1)(s-2)(s+1)} = -1$$

$$B = \lim_{s \rightarrow 1} \frac{6 + (s+1)(s^2 - 5s + 11)}{s(s-1)(s-2)(s+1)} = -10$$

$$C = \lim_{s \rightarrow 2} \frac{6 + (s+1)(s^2 - 5s + 11)}{s(s-1)(s-2)(s+1)} = \frac{7}{2}$$

$$f(t) = \left[\frac{17}{2} - 10e^t + \frac{7}{2}e^{2t} - e^{-t} \right] u(t)$$